

1. BIT STUFFING

Server code

```
import socket

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

server_socket.bind(("localhost",18000))

server_socket.listen(1)

print("server is waiting")

conn,addr=server_socket.accept()

print("connected with",addr)

data=conn.recv(1024).decode()

count=0

res=""

for bit in data:

    if bit=='1':

        count=count+1

        res=res+'1'

        if count==5:

            res=res+'0'

            count=0

    else:

        res=res+'0'

        count=0

conn.send(res.encode())

conn.close()

server_socket.close()
```

Client code

```
import socket

client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

client_socket.connect(("localhost",18000))

data=input("Enter bit structure:")

client_socket.send(data.encode())

stuffed=client_socket.recv(1024).decode()

print("Stuffed bits:",stuffed)
```

2. BYTE STUFFING

Server code

```
import socket

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

server_socket.bind(("localhost",17000))

server_socket.listen(1)

print("Server is waiting for connection")

conn,addr=server_socket.accept()

data=conn.recv(1024).decode()

res="F"

for ch in data:

    if ch=='F' or ch=='E':

        res=res+'E'

    res=res+ch

res=res+'F'

conn.send(res.encode())
```

Client code

```
import socket

client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

client_socket.connect(("localhost",17000))

data=input("Enter data:")

client_socket.send(data.encode())

stuffed=client_socket.recv(1024).decode()

print("Stuffed data:",stuffed)
```

3. CRC

Server code

```
import socket

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

server_socket.bind(("localhost",4446))

server_socket.listen(1)

print("server is waiting for connection....")

conn,addr=server_socket.accept()

print("connection with",addr)

data=conn.recv(1024).decode()

div=conn.recv(1024).decode()

data=data+"0"*(len(div)-1)

data=list(data)

div=list(div)

for i in range(len(data)-len(div)+1):

    if data[i]=='1':

        for j in range(len(div)):

            if data[i+j]==div[j]:

                data[i+j]='0'
```

```

        else:
            data[i+j]='1'
rem=""
for i in range(len(div)-1):
    rem=rem+data[-(len(div)-1)+1]
conn.send(rem.encode())

```

Client code

```

import socket

client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
client_socket.connect(("localhost",4446))
data=input("enter the data:")
client_socket.send(data.encode())
div=input("enter the divisor")
client_socket.send(div.encode())
rem=client_socket.recv(1024).decode()
print(rem)

```

4. PALLINDROME

Server code

```

import socket

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
server_socket.bind(("localhost",18000))
server_socket.listen(1)
print("server is waiting for connection....")
conn,addr=server_socket.accept()
print("connection with",addr)
data=conn.recv(1024).decode()

```

```
if data==data[::-1]:  
    result="it is pallin"  
else:  
    result="it is not pallin"  
conn.send(str(result).encode())  
conn.close()  
server.socket.close()
```

Client code

```
import socket  
client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)  
client_socket.connect(("localhost",18000))  
text=input("enter the text:")  
client_socket.send(text.encode())  
result=client_socket.recv(1024).decode()  
print("Server response:",result)  
client_socket.close()
```

5. FACTORIAL

Server code

```
import socket  
import math  
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)  
host = "localhost"  
port = 12345  
server.bind((host, port))  
server.listen(1)
```

```
print("Server waiting for connection...")
conn, addr = server.accept()
print("Connected by:", addr)
number = int(conn.recv(1024).decode())
print("Received number:", number)
factorial = math.factorial(number)
conn.send(str(factorial).encode())
conn.close()
server.close()
```

Client code

```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
host = "localhost"
port = 12345
client.connect((host, port))
num = input("Enter a number: ")
client.send(num.encode())
result = client.recv(1024).decode()
print("Factorial received from server:", result)
client.close()
```

6.FIBINACCI SERIES

Server code

```
import socket
server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
server_socket.bind(("localhost",4446))
server_socket.listen(1)
```

```
print("server is waiting for connection....")
conn,addr=server_socket.accept()
print("connection with",addr)
data=conn.recv(1024).decode()
def fibonacci(n):
    series=[]
    a,b=0,1
    for i in range(n):
        series.append(a)
        a,b=b,a+b
    return series
n=int(data)
result=fibonacci(n)
conn.send(str(result).encode())
conn.close()
server_socket.close()
```

Client code

```
import socket
client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
client_socket.connect(("localhost",4446))
n=input("enter the data:")
client_socket.send(n.encode())
data=client_socket.recv(1024).decode()
print("fib series:",data)
client_socket.close()
```

7. FTP

Server code

```
import socket

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

server_socket.bind(("localhost",18000))

server_socket.listen(1)

print("server is waiting for connection....")

conn,addr=server_socket.accept()

print("connection with",addr)

data=conn.recv(1024)

f=open("recv.txt","wb")

while data:

    f.write(data)

    data=conn.recv(1024)

f.close()

conn.close()

server_socket.close()

print("File received")
```

Client code

```
import socket

client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

client_socket.connect(("localhost",18000))

f=open("send.txt","rb")

data=f.read(1024)

while data:

    client_socket.send(data)

    data=f.read(1024)
```



```
f.close()
client_socket.close()
print("File sent")
```

create a new file named send.txt and store some sentence and create another file names recv.txt and store nothing in it

8.Database

Server code

```
import socket
import sqlite3

server_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)
server_socket.bind(("localhost",5500))
server_socket.listen(1)
print("Server is waiting...")
conn,addr=server_socket.accept()
query=conn.recv(1024).decode()
db=sqlite3.connect("test.db")
cur=db.cursor()
cur.execute(query)
rows=cur.fetchall()
result=str(rows)
conn.send(result.encode())
db.close()
conn.close()
server_socket.close()
```

Client code

```
import socket

client_socket=socket.socket(socket.AF_INET,socket.SOCK_STREAM)

client_socket.connect(("localhost",5500))

query=input("Enter query:")

client_socket.send(query.encode())

data=client_socket.recv(1024).decode()

print("Result:",data)

client_socket.close()
```

db.py

```
import socket

import sqlite3

conn=sqlite3.connect("test.db")

cur=conn.cursor()

cur.execute("create table if not exists students(id int,name text)")

cur.execute("insert into students values(1,'Nets')")

cur.execute("insert into students values(2,'Nea')")

conn.commit()

conn.close()
```

create a new file named test.db and store nothing in it